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ELECRO-STATIC CHUCK AND SUBSTRATE PROCESSING APPARATUS INCLUDING THE SAME

Abstract

An electro-static chuck includes a chuck plate configured to seat a substrate, an insulating pillar on the outside of the chuck plate, a first ring surrounding a side portion of the chuck plate on the insulating pillar, a second ring covering at least a portion of an upper portion of the first ring, and a heating plate configured to discharge heat from a lower portion of the chuck plate to heat the substrate, wherein the second ring is arranged such that an overlap of a first region of the substrate affected by the heating plate and a second region of the substrate affected by the second ring is minimized.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Korean Patent Application No. 10-2024-0030110, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] An electro-static chuck may include a heating plate for heating a substrate, and a ring surrounding the substrate. The ring may determine the shape of plasma formed in a chamber of a substrate processing apparatus. Based on the position of the ring, the plasma shape may be changed.

SUMMARY

[0003] In general, in some aspects, the present disclosure is directed toward an electro-static chuck having improved reliability and a substrate processing apparatus including the electro-static chuck.

[0004] According to some implementations, the present disclosure is directed to an electro-static chuck that includes a chuck plate configured to seat a substrate, an insulating pillar on the outside of the chuck plate, a first ring surrounding a side portion of the chuck plate on the insulating pillar, a second ring covering at least a portion of an upper portion of the first ring, and a heating plate configured to discharge heat from a lower portion of the chuck plate to heat the substrate, wherein the second ring is arranged such that an overlap of a first region of the substrate affected by the heating plate and a second region of the substrate affected by the second ring is minimized.

[0005] According to some implementations, the present disclosure is directed to an electro-static chuck that includes a chuck plate configured to seat a substrate, an insulating pillar which is formed on the outside of the chuck plate and has a pin hole, a first ring surrounding a side portion of the chuck plate on the insulating pillar, a second ring covering at least a portion of an upper portion of the first ring, a driving pin configured to be movable in a vertical direction from the pin hole of the insulating pillar and overlapping at least a portion of the second ring in the vertical direction, and a heating plate configured to discharge heat from a lower portion of the chuck plate to heat the substrate, wherein the second ring is arranged such that an overlap of a first region of the substrate affected by the heating plate and a second region of the substrate affected by the second ring is minimized.

[0006] According to some implementations, the present disclosure is directed to an electro-static chuck that includes a chuck plate configured to seat a substrate, an insulating pillar on the outside of the chuck plate, a first ring surrounding a side portion of the chuck plate on the insulating pillar, a second ring covering at least a portion of an upper portion of the first ring, and a heating plate configured to discharge heat from a lower portion of the chuck plate to heat the substrate, wherein the chuck plate includes a first portion on which the substrate is seated and a second portion extending from a lower portion of the first portion to the outside of the first portion, and the second ring is spaced apart from the first portion in a horizontal direction by about 3.5 mm or more.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Example implementations will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings.

[0008] FIG. **1** is a cross-sectional view showing an example of a electro-static chuck according to some implementations.

[0009] FIG. **2** is an enlarged cross-sectional view of an example of a side portion of the electro-static chuck of FIG. **1** according to some implementations.

[0010] FIG. **3** is a plan view illustrating an example of a region of a substrate according to some implementations

[0011] FIG. **4** is a plan view illustrating an example of a region of a substrate according to some implementations.

[0012] FIG. **5** is a plan view illustrating an example of a region of a substrate according to some implementations

[0013] FIG. **6** is a cross-sectional view showing an example of an electro-static chuck according to some implementations.

[0014] FIG. **7** is an enlarged cross-sectional view of an example of a side portion of the electro-static chuck of FIG. **6** according to some implementations.

[0015] FIG. **8** is an enlarged cross-sectional view of an example of a side portion of the electro-static chuck of FIG. **6** according to some implementations.

[0016] FIG. **9** is a cross-sectional view illustrating an example of a substrate processing apparatus according to some implementations.

DETAILED DESCRIPTION

[0017] Hereinafter, example implementations will be explained in detail with reference to the accompanying drawings. The same reference numerals are used for the same elements in the drawings, and redundant descriptions thereof are omitted.

[0018] FIG. **1** is a cross-sectional view showing an example of an electro-static chuck according to some implementations, and FIG. **2** is an enlarged cross-sectional view of an example of a side portion of the electro-static chuck of FIG. **1** according to some implementations. FIG. **3** is a plan view illustrating an example of a region of a substrate according to some implementations. FIG. **2** illustrates a side portion A_**1** of an electro-static chuck **10** from which a driving pin **160** (of FIG. **6**) is omitted.

[0019] In FIGS. **1** to **3**, the electro-static chuck **10** may include a body portion **110**, a chuck plate **120**, an insulating pillar **130**, a first ring **140**, and a second ring **150**. In addition, the electro-static chuck **10** may further include an electro-static plate **210**, a heating plate **220**, and a cooling plate **230**.

[0020] In some implementations, the body portion **110** may be a pedestal having a cylindrical shape. The body portion **110** may accommodate the electro-static plate **210**, the heating plate **220**, the cooling plate **230**, and the like therein.

[0021] The body portion **110** may include a first portion **111** on which the chuck plate **120** is seated and a second portion **112** extending from a lower portion of the first portion **111** to the outside of the first portion **111**. When the body portion **110** is viewed from top to bottom, the top surface of the first portion **111** and the top surface of the second portion **112** may be partially exposed.

[0022] In some implementations, the chuck plate **120** may be on the body portion **110**. The chuck plate **120** may be a plate on which a substrate S is seated. For example, the substrate S seated on the chuck plate **120** may be a wafer on which semiconductor devices are formed. The chuck plate **120** may include a non-conductive material having little deformation by heat for resistance by high-temperature plasma.

[0023] Here, the diameter of the substrate S may be about 300 mm, but is not limited thereto. The diameter of the substrate S may be, for example, about 150 mm, about 200 mm, or about 450 mm, or more. Hereinafter, a case in which the diameter of the substrate S is about 300 mm will be described as an example.

[0024] The chuck plate **120** may include a first portion **121** on which the substrate S is seated and a second portion **122** extending from a lower portion of the first portion **121** to the outside of the first portion **121**. When the chuck plate **120** is viewed from top to bottom, the top surface of the first portion **121** and the top surface of the second portion **122** may be partially exposed.

[0025] In some implementations, the insulating pillar **130** may be at the outside the chuck plate **120**. More specifically, the insulating pillar **130** may surround the body portion **110** from the outside of the chuck plate **120**. In addition, the insulating pillar **130** may be a pillar including an insulating material.

[0026] The first ring **140** may be a ring surrounding a side portion of the chuck plate **120** on the insulating pillar **130**. More specifically, the first ring **140** may surround the side surface of the first portion **121** and the top surface of the second portion **122** of the chuck plate **120** on the insulating pillar **130**. Since the first ring **140** surrounds the side portion of the chuck plate **120**, the risk of damage to the chuck plate **120** by plasma in a substrate processing process may be reduced. The first ring **140** is a ring for protecting the chuck plate **120** and may be referred to as a protection ring and/or an insert ring. In addition, when the substrate S is seated on the chuck plate **120**, a part of the first ring **140** may overlap a part of the edge of the substrate S in a vertical direction (Z direction).

[0027] In the present disclosure, a direction parallel to the main surface of the substrate S may be defined as a horizontal direction (X direction and/or Y direction), and a direction perpendicular to the horizontal direction (X direction and/or Y direction) may be defined as a vertical direction (Z direction).

[0028] The first ring **140** may include a support portion **141** and a protection portion **142**. In some implementations, the support portion **141** may be a portion of the first ring **140** seated on the insulating pillar **130**. The support portion **141** may be in contact with the insulating pillar **130**.

[0029] In some implementations, the protection portion **142** may be a portion of the first ring **140** extending in the horizontal direction (X and/or Y directions) from the inside of the support portion **141** and surrounding the side portion of the chuck plate **120**. More specifically, the protection portion **142** may extend in the horizontal direction (X and/or Y directions) from the inside of the support portion **141** to surround the side surface of the first portion **121** and the top surface of the second portion **122** of the chuck plate **120**. The protection portion **142** may be in contact with the side portion of the chuck plate **120**.

[0030] The second ring **150** may be seated on the insulating pillar **130** and may partially cover the upper portion of the insulating pillar **130** and the upper portion of the first ring **140**. The second ring **150** may prevent etching of the first ring **140** by plasma in a substrate processing process.

[0031] The second ring **150** may be a ring that affects a shape of plasma formed in a substrate processing process. For example, when the shape of the second ring **150** or the separation distance in the vertical direction (Z direction) of the second ring **150** and the insulating pillar **130** changes, the shape of plasma formed in the substrate processing process may also change. For example, when the shape of the second ring **150** or the separation distance between the second ring **150** and the insulating pillar **130** changes, the shape of the plasma inside a process chamber **1501** of a substrate processing apparatus **1** (FIG. **9**) may change. The second ring **150** may be referred to as a focus ring and/or an edge ring. In addition, the second ring **150** may be a ring including materials, such as quartz, silicon, silicon carbide, silicon oxide, aluminum oxide, etc.

[0032] The second ring **150** may include a contact portion **151** and a cover portion **152**. The contact portion **151** may be a portion of the second ring **150** surrounding a side portion of the first ring **140** on the insulating pillar **130**. The contact portion **151** may be in contact with each of the insulating pillar **130** and the first ring **140**.

[0033] The cover portion **152** may be a portion of the second ring **150** that extends in the horizontal direction (X and/or Y directions) on the contact portion **151** and partially covers the upper portion of the first ring **140**. More specifically, the cover portion **152** may extend in the horizontal direction (X direction and/or Y direction) from the upper portion of the contact portion **151** and may partially cover the first ring **140**.

[0034] The top surface of the second ring **150** may be positioned at a vertical level higher than the top surface of the chuck plate **120**. The top surface of the cover portion **152** may be positioned at a vertical level higher than the top surface of the chuck plate **120**. In addition, the top surface of the second ring **150** may be located at a vertical level higher than the top surface of a substrate S loaded on the chuck plate **120**.

[0035] Although FIGS. **1** and **2** show that the electro-static chuck **10** includes two rings, some implementations are not limited thereto. For example, the electro-static chuck **10** may include one ring or three or more rings.

[0036] In some implementations, the electro-static plate **210** may be a plate that generates an electro-static force under the chuck plate **120**. The electro-static plate **210** may be electrically connected to an electro-static chuck power device (**1530** of FIG. **9**) to be described below. An electro-static force may be generated between the electro-static plate **210** and the substrate S by the power applied from the electro-static chuck power device **1530**, for example, a direct current voltage. The substrate S may be firmly seated on the chuck plate **120** by the electro-static force.

[0037] In some implementations, the heating plate **220** may be a plate configured to dissipate heat to heat the substrate S on the chuck plate **120**. The heating plate **220** may be electrically connected to a heater power device (**1550** of FIG. **9**) to be described later. The heating plate **220** may include a plurality of heating elements. For example, the heating plate **220** may include at least one of a thermoelectric element, a resistance heater, and an inductance heater. The plurality of heating elements may be individually controlled for local temperature control of the substrate S on the chuck plate **120**.

[0038] In some implementations, the cooling plate **230** may be a plate configured to cool a substrate on the chuck plate **120** or a plurality of electronic devices included in the electro-static chuck **10**. The cooling plate **230** may be electrically connected to a temperature control device (**1570** of FIG. **9**) to be described later. The cooling plate **230** may include a cooling water channel **231** through which at least one cooling water among water, ethylene glycol, and silicone oil flows.

[0039] The substrate S may include a plurality of regions. For example, the substrate S may include a first region A**1** and a second region A**2**. When viewed in a plan view, the second region A**2** may surround the first region A**1**. For example, the first region A**1** may refer to the region of the substrate S affected (i.e., heated) by the heating plate **220**, and the second region A**2** may refer to the region of the substrate S affected by the second ring **150**. When viewed in a plan view, the first region A**1** may be arranged adjacent to the center C of the substrate S, and the second region A**2** may be arranged adjacent to the edge of the substrate S.

[0040] When viewed in a plan view, the first region A**1** may have a circular shape, and the second region A**2** may have a ring shape. However, some implementations are not limited thereto, and the shape of each of the first region A**1** and/or the second region A**2** may be variously modified.

[0041] Hereinafter, a case in which the first region A**1** has a circular shape and the second region A**2** has a ring shape will be described as an example. Accordingly, the first region A**1** may have a radius, and the second region A**2** may have an inner diameter ID and an outer diameter OD.

[0042] The first region A**1** may have a first width W**1** in the radial direction of the substrate S at the center C of the substrate S, and the second region A**2** may have a second width W**2** in the radial direction of the substrate S. The second width W**2** may be a thickness of the second region A**2** extending in a radial direction of the substrate S of the second region A**2**. That is, the second width W**2** may be a thickness of a ring-shaped region.

[0043] For example, the first width W**1** may be about 120 mm to about 147 mm, and the second width W**2** may be about 3 mm to about 15 mm. For example, the center of the first region A**1** and

the center C of the substrate S may coincide with each other. In some implementations, the center of the first region A**1** and the center C of the substrate S may be spaced apart from each other.
[0044] For efficient control of the substrate S, the second ring **150** may be arranged so that an overlap between the first region A**1** and the second region A**2** is minimized. When the first region A**1** and the second region A**2** overlap, the frequency of occurrence of discontinuities of process parameters increases, process distribution increases in plasma substrate processing, and thus the reliability of the plasma processing process may decrease. Accordingly, when the overlap between the first region A**1** and the second region A**2** is minimized, the reliability of the plasma substrate processing process may increase.
[0045] For example, the second ring **150** may be arranged so that the first region A**1** and the second region A**2** do not overlap each other. For example, the heating plate **220** and the second ring **150** may be arranged so that the first region A**1** and the second region A**2** do not overlap each other and the sum of the plane area of the first region A**1** and the plane area of the second region A**2** are the same as the plane area of the substrate S. That is, the first region A**1** and the second region A**2** may be in contact with each other. That is, the sum of the first width W**1** and the second width W**2** may be the same as the radius of the substrate S.
[0046] For example, the second ring **150** may be arranged with a horizontal distance L from the substrate S in the horizontal direction (X and/or Y directions). As the horizontal distance L between the second ring **150** and the substrate S increases, the plane area of the second region A**2** decreases, and the horizontal separation distance between the second region A**2** and the center C of the substrate S may increase. In more detail, as the horizontal distance L between the second ring **150** and the substrate S increases, the inner diameter ID and/or the outer diameter OD of the second region A**2** may increase. In some implementations, the outer diameter OD of the second region A**2** may be the same as the radius of the substrate S. In some implementations, the outer diameter OD of the second region A**2** may be less than the radius of the substrate S. That is, as the horizontal distance L between the second ring **150** and the substrate S increases, the overlapping region of the first region A**1** and the second region A**2** may decrease.
[0047] For example, the horizontal distance L may be greater than or equal to about 3.5 mm. The horizontal distance L may be variously modified by the radius of the substrate S, the arrangement of the heating plate **220**, and/or the shape of the body portion **110**. For example, the second ring **150** may be spaced apart from the first portion **121** of the chuck plate **120** by about 3.5 mm or more in the horizontal direction (X and/or Y directions).
[0048] In general, an electro-static chuck may include a second ring is arranged without considering the overlap between the first region and the second region of the substrate. Accordingly, the overlapping region of the first region and the second region on the substrate was wide. A distribution inflection point occurred in a region where the first region and the second region overlap, and the reliability of the plasma processing process was low.
[0049] In the electro-static chuck **10**, the second ring **150** may be arranged so that an overlap between the first region A**1** and the second region A**2** of the substrate S is minimized. Accordingly, the region in which the distribution inflection point on the substrate S occurs may be minimized, and the reliability of the plasma processing process may be increased.
[0050] FIGS. **4** and **5** are plan views illustrating an example of a region of a substrate according to some implementations. FIG. **4** illustrates a case where a first region Ala and a second region A**2***a* do not overlap each other, and the sum of the plane area of the first region Ala and the plane area of the second region A**2***a* is less than the plane area of the substrate S. FIG. **5** illustrates a case where at least a portion of each of the first region Alb and the second region A**2***b* overlaps. Description will be made with reference to FIGS. **4** and **5** together with FIGS. **1** to **3**.
[0051] In FIGS. **4** and **5**, the first region Ala of the substrate S of FIG. **4** may have a first width W**1***a*, and the second region A**2***a* thereof may have a second width W**2***a*. In addition, the first region Alb of the substrate S of FIG. **5** may have a first width W**1***b*, and the second region A**2***b* may

have a second width W**2***b*. The arrangement positions of the second ring **150** and/or the heating plate **220** may be adjusted so that the first widths W**1***a* and W**1***b* and/or the second widths W**2***a* and W**2***b* may be modified.

[0052] In some implementations, each of the first width W**1***a* of FIG. **4** and the first width W**1***b* of FIG. **5** may be the same as the first width W**1** of FIG. **3**. In some implementations, the first width W**1***a* of FIG. **4** and/or the first width W**1***b* of FIG. **5** may be different from the first width W**1** of FIG. **3**. Also, in some implementations, each of the second width W**2***a* of FIG. **4** and the second width W**2***b* of FIG. **5** may be different from the second width W**2** of FIG. **3**.

[0053] The sum of the plane area of the first region Ala of FIG. **4** and the plane area of the second region A**2***a* may be less than the plane area of the substrate S. The first region Ala and the second region A**2***a* of FIG. **4** may be spaced apart from each other in the horizontal direction (X direction and/or Y direction). In addition, the sum of the plane area of the first region Alb and the plane area of the second region A**2***b* of FIG. **5** may be greater than the plane area of the substrate S. That is, at least a portion of each of the first region Alb and the second region A**2***b* of FIG. **5** may overlap each other.

[0054] That is, the sum of the first width W**1***a* and the second width W**2***a* of FIG. **4** may be less than the radius of the substrate S. In addition, the sum of the first width W**1***b* and the second width W**2***b* of FIG. **5** may be greater than the radius of the substrate S.

[0055] FIG. **6** is a cross-sectional view showing an example of an electro-static chuck according to some implementations, and FIGS. **7** and **8** are enlarged cross-sectional views of an example of a side portion of the electro-static chuck of FIG. **6** according to some implementations. FIG. **7** shows a side portion A_II of the electro-static chuck **10***a* in which a driving pin **160** is in a first state, and FIG. **8** shows a side portion A_III of the electro-static chuck **10***a* in which the driving pin **160** is in a second state.

[0056] In FIGS. **6** to **8**, the electro-static chuck **10***a* may include the body portion **110**, the chuck plate **120**, the insulating pillar **130***a*, the first ring **140**, the second ring **150**, a driving pin **160**, and a power source **170**. In addition, the electro-static chuck **10***a* may further include the electro-static plate **210**, the heating plate **220**, and the cooling plate **230**. The electro-static chuck **10***a* of FIGS. **6** to **8** may further include the driving pin **160** and the power source **170** in the electro-static chuck **10** of FIGS. **1** and **2**.

[0057] The insulating pillar **130***a* may have a pin hole H formed therein. The pin hole H may provide a space in which the driving pin **160** may move in the vertical direction (Z direction). The pin hole H may overlap at least a portion of the second ring **150** in the vertical direction (Z direction). In some implementations, the pin hole H may overlap at least a portion of each of the first ring **140** and the second ring **150** in the vertical direction (Z direction).

[0058] The second ring **150** may be driven in the vertical direction (Z direction) by the driving pin **160**. In the substrate processing process, the second ring **150** may move in the vertical direction (Z direction) in order to implement the plasma in a previously predicted shape. More specifically, the driving pin **160** may be under the second ring **150**, and at least a portion of the second ring **150** may overlap a portion of the driving pin **160** in the vertical direction (Z direction). The second ring **150** may be driven in the vertical direction (Z direction) by an external force transmitted from the driving pin **160** to the second ring **150**.

[0059] In addition, in some implementations, the second ring **150** may be driven in the vertical direction (Z direction) while the substrate S is seated on the chuck plate **120**. The second ring **150** may be moved in the vertical direction (Z direction) in the process of processing the substrate S. The separation distance between the second ring **150** and the insulating pillar **130***a* in the vertical direction (Z direction) may be controlled by the driving pin **160**. Accordingly, the shape of the plasma formed in the substrate processing process may also be controlled by the driving pin **160**. For example, when the shape of the plasma in the substrate processing process is different from the shape predicted in advance, the second ring **150** may be driven in the vertical direction (Z direction) by the driving pin **160**. Accordingly, in the substrate processing process, plasma may be formed in a previously predicted shape.

[0060] In some implementations, the driving pin **160** may be a pin that moves in the vertical direction (Z direction) in the pin hole H of the insulating pillar **130***a*. More specifically, the driving pin **160** may move in the vertical direction (Z direction) within the pin hole H to drive the second ring **150** in the vertical direction (Z direction).

[0061] The driving pin **160** may include a rod-shaped pin extending in the vertical direction (Z direction) in the pin hole H. There may be a plurality of driving pins **160**, and the plurality of driving pins **160** may be formed to be symmetrical with respect to the center of the chuck plate **120**. For example, there may be three driving pins **160**, and the three driving pins **160** may be formed symmetrically with respect to the center of the chuck plate **120**.

[0062] The driving pin **160** may overlap at least a portion of the second ring **150** in the vertical direction (Z direction). Accordingly, the driving pin **160** may drive the second ring **150** in the vertical direction (Z direction).

[0063] The driving pin **160** may be in one of the first state and the second state. The first state of the driving pin **160** may be a state in which the driving pin **160** does not drive the second ring **150** in the vertical direction (Z direction). In the first state, the driving pin **160** may be in contact with the second ring **150**. In some implementations, in the first state, the driving pin **160** may be spaced apart from the second ring **150** in the vertical direction (Z direction).

[0064] The second state of the driving pin **160** may be a state in which the driving pin **160** moves to drive the second ring **150** in the vertical direction (Z direction). As described above, the driving pin **160** may drive the second ring **150** in the vertical direction (Z direction) to change the shape of plasma formed in the substrate processing process. When the driving pin **160** is in the second state, the second ring **150** may be driven in the vertical direction (Z direction) to be spaced apart from the insulating pillar **130** in the vertical direction (Z direction). When the driving pin **160** is in the second state, the driving pin **160** may not drive the first ring **140** in the vertical direction (Z direction). In some implementations, when the driving pin **160** is in the second state, the driving pin **160** may drive the first ring **140** in the vertical direction (Z direction).

[0065] In some implementations, it is not limited to those illustrated in FIGS. **6** to **8**, and the electro-static chuck **10***a* may include three or more rings. In addition, the plurality of rings may overlap at least a portion of the driving pin **160** in the vertical direction (Z direction). Accordingly, when the driving pin **160** moves in the vertical direction (Z direction), at least one of the plurality of rings may be driven in the vertical direction (Z direction).

[0066] In some implementations, the power source **170** may be a device that transmits power to the driving pin **160**. More specifically, the power source **170** may be a device that transmits power to the driving pin **160** for movement of the driving pin **160** in the vertical direction (Z direction). For example, the power source **170** may include a hydraulic device, a motor, and the like.

[0067] The electro-static chuck **10***a* may further include a control unit for controlling the operations of the components of the electro-static chuck **10***a*. The control unit may control movement of the driving pin **160** in the vertical direction (Z direction) by controlling the power source **170**. That is, the control unit may control movement of the second ring **150** in the vertical direction (Z direction). The control unit may control the movement of the second ring **150** in the vertical direction (Z direction) by controlling the power source **170** to control the plasma on the substrate S.

[0068] The control unit may be implemented in hardware, firmware, software, or any combination thereof. For example, the control unit may be a computing device, such as a workstation computer, a desktop computer, a laptop computer, a tablet computer, or the like. For example, the control unit may include memory devices, such as Read Only Memory (ROM) and Random Access Memory (RAM), and a processor configured to perform predetermined operations and algorithms, such as a microprocessor, a central processing unit (CPU), and a graphics processing unit (GPU). In addition, the control unit may include a receiver and a transmitter for receiving and transmitting electrical signals.

[0069] FIG. **9** is a cross-sectional view illustrating an example of a substrate processing apparatus according to some implementations. The substrate processing apparatus may be a substrate processing apparatus including at least one of the electro-static chuck **10** and **10***a* described above. Description will be made with reference to FIG. **9** together with FIGS. **1** to **8**.

[0070] In FIG. **9**, the substrate processing apparatus **1** may include an electro-static chuck **10**, a process chamber **1501**, a gas supply pipe **1502**, a plasma generating device **1503**, a gate **1504**, a pump **1505**, a high-frequency power device **1510**, a gas supply device **1520**, an electro-static chuck power device **1530**, a heater power device **1540**, a control unit **1550**, a bias power device **1560**, a temperature control apparatus **1570**, and the like.

[0071] In some implementations, the electro-static chuck **10** of the substrate processing apparatus **1** may be a device that fixes the substrate S by electro-static force. Since the technical idea of the electro-static chuck **10** may be substantially the same as those described with reference to FIGS. **1** to **8**, detailed descriptions thereof will be omitted.

[0072] In some implementations, the process chamber **1501** may provide an inner space for processing the substrate S. The electro-static chuck **10** may be positioned in the inner space of the process chamber **1501**. The gas supply pipe **1502** may be connected to the gas supply device **1520**. The gas supply pipe **1502** may be configured to inject the process gas provided by the gas supply device **1520** into the process chamber **1501**. The processing gas may include an etching gas for etching the substrate S. In addition, the processing gas may include a protection gas for protecting patterns formed on the substrate S.

[0073] In some implementations, the gate **1504** may provide a path through which the substrate S may move. For example, the substrate S may move to the outside of the process chamber **1501** through the gate **1504**, or may move to the inside of the process chamber **1501** through the gate **1504**. The pump **1505** may be configured to adjust an internal pressure of the process chamber **1501**. For example, the pump **1505** may increase the pressure by injecting air into the process chamber **1501**. In addition, the pump **1505** may discharge air inside the process chamber **1501** to reduce pressure.

[0074] In some implementations, the high-frequency power device **1510** may be electrically connected to the plasma generating device **1503**. The high-frequency power device **1510** may output high-frequency power suitable for generating plasma and transmit the output high-frequency power to the plasma generating device **1503**. The high-frequency power of the high-frequency power device **1510** may be controlled by the control unit **1550**.

[0075] In some implementations, the electro-static chuck power device **1530** may be electrically connected to the electro-static chuck **10**. More specifically, the electro-static chuck power device **1530** may be electrically connected to the electro-static plate **210** of the electro-static chuck **10**. An electro-static force may be generated between the electro-static plate **210** and the substrate S by the power applied from the electro-static chuck power device **1530**, for example, a direct current voltage. The substrate S may be firmly seated on the chuck plate **120** of the electro-static chuck **10** by the electro-static force.

[0076] In some implementations, the heater power device **1540** may be electrically connected to the heating plate **220**. The heater power device **1540** may be connected to the control unit **1550**, and the heating amount of a plurality of heating elements included in the heating plate **220** may be controlled.

[0077] In some implementations, the bias power device **1560** may be connected to a lower portion of the body portion **110**. The bias power device **1560** may apply high-frequency power to a lower portion of the body **110**. The lower portion of the body portion **110** may serve as an electrode for generating plasma.

[0078] In some implementations, the temperature control device **1570** may be connected to the cooling water channel **231** of the cooling plate **230** and the control unit **1550**. The temperature control device **1570** may control the temperature of the cooling water flowing through the cooling water channel **231**.

[0079] In some implementations, the control unit **1550** may be configured to control at least one of the high-frequency power device **1510**, the gas supply device **1520**, the electro-static chuck power device **1530**, the heater power device **1540**, the bias power device **1560**, and the temperature control device **1570**.

[0080] In some implementations, the control unit **1550** may be configured to control the power source **170** of the electro-static chuck **10**. The power source **170** may be controlled by the control unit **1550** to drive the second ring **150**. The electro-static chuck **10** of the embodiments may be in at least one of the first state and the second state described above by the power source **170** controlled by the control unit **1550**.

[0081] While this disclosure contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed. Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a combination can in some cases be excised from the combination, and the combination may be directed to a subcombination or variation of a subcombination.

Claims

1. An electro-static chuck comprising: a chuck plate configured to seat a substrate; an insulating pillar on an exterior of the chuck plate; a first ring surrounding a side portion of the chuck plate on the insulating pillar; a second ring covering at least a portion of an upper portion of the first ring; and a heating plate configured to transmit heat from a lower portion of the chuck plate to heat the substrate, wherein the second ring is arranged such that an overlap of a first region of the substrate affected by the heating plate and a second region of the substrate affected by the second ring is minimized.

2. The electro-static chuck of claim 1, wherein, from a plan view, the first region of the substrate and the second region of the substrate are adjacent to each other.

3. The electro-static chuck of claim 1, wherein, from a plan view, the second region of the substrate surrounds the first region of the substrate.

4. The electro-static chuck of claim 1, wherein, from a plan view, the first region of the substrate is placed adjacent to a center of the substrate, and the second region of the substrate is placed adjacent to an edge of the substrate.

5. The electro-static chuck of claim 1, further comprising a body portion including a first portion upon which the chuck plate is seated and a second portion extending from a lower portion of the first portion to an exterior of the first portion, wherein the heating plate is placed inside the body portion.

6. The electro-static chuck of claim 1, wherein the second ring comprises: a contact portion surrounding a side portion of the first ring on the insulating pillar; and a cover portion extending in a horizontal direction and surrounding an upper portion of the first ring on the contact portion.

7. The electro-static chuck of claim 1, wherein the first region of the substrate has a first width in a radial direction of the substrate from a center of the substrate, and the second region of the substrate has a second width in the radial direction of the substrate, wherein the first width is within a range of about 120 mm to about 147 mm, and wherein the second width is within a range of about 3 mm to about 15 mm.

8. The electro-static chuck of claim 1, wherein a horizontal separation distance between the substrate and the second ring is about 3.5 mm or more.

9. An electro-static chuck comprising: a chuck plate configured to seat a substrate; an insulating pillar formed on an exterior of the chuck plate, the insulating pillar having a pin hole; a first ring surrounding a side portion of the chuck plate on the insulating pillar; a second ring covering at least a portion of an upper portion of the first ring; a driving pin configured to be movable in a vertical direction withing the pin hole of the insulating pillar and overlapping at least a portion of the second ring in the vertical direction; and a heating plate configured to transmit heat from a lower portion of the chuck plate to heat the substrate, wherein the second ring is arranged such that an overlap of a first region of the substrate affected by the heating plate and a second region of the substrate affected by the second ring is minimized.

10. The electro-static chuck of claim 9, wherein the driving pin is configured to drive the second ring in the vertical direction.

11. The electro-static chuck of claim 9, further comprising a power source configured to transmit power to the driving pin.

12. The electro-static chuck of claim 9, wherein the first ring is spaced apart from the driving pin in the vertical direction.

13. The electro-static chuck of claim 9, wherein, from a plan view, the first region has a circular shape and the second region has a ring shape.

14. The electro-static chuck of claim 13, wherein, from a plan view, an outer diameter of the second region is the same as an outer diameter of the substrate.

15. The electro-static chuck of claim 9, wherein a sum of a plane area of the first region and a plane area of the second region is equal to or smaller than a plane area of the substrate.

16. An electro-static chuck comprising: a chuck plate configured to seat a substrate; an insulating pillar on an exterior of the chuck plate; a first ring surrounding a side portion of the chuck plate on the insulating pillar; a second ring covering at least a portion of an upper portion of the first ring; and a heating plate configured to transmit heat from a lower portion of the chuck plate to heat the substrate, wherein the chuck plate includes a first portion on which the substrate is seated and a second portion extending from a lower portion of the first portion to an exterior of the first portion, and wherein the second ring is spaced apart from the first portion in a horizontal direction by about 3.5 mm or more.

17. The electro-static chuck of claim 16, wherein the insulating pillar further comprises: a pin hole; and a driving pin configured to be movable in a vertical direction within the pin hole of the insulating pillar and overlapping at least a portion of the second ring in the vertical direction.

18. The electro-static chuck of claim 16, wherein, from a plan view, a first region of the substrate has a circular shape and a second region of the substrate has a ring shape, and wherein a sum of a radius of the first region and a width from an inner diameter to an outer diameter of the second region is the same as a radius of the substrate or less than a radius of the substrate.

19. The electro-static chuck of claim 18, wherein the first region of the substrate and the second region of the substrate are in contact with each other.

20. The electro-static chuck of claim 16, wherein the second ring comprises at least one of quartz, silicon, silicon carbide, silicon oxide, and aluminum oxide.